**Listing 12.6 Matching restaurant locations in a georectangle**

```
GET restaurants/_search
{
  "query": {
    "geo_bounding_box": {
      "location": {
        "top_left": {
          "lat": 52,
          "lon": 0.2
        },
        "bottom_right": {
          "lat": 49,
          "lon": 0.1
        }
      }
    }
  }
}
```

Constructs a georectangle

Sets the document's geo_point field

Defines the top_left point, formed by a latitude/longitude pair

Defines the bottom_right point, formed by a latitude/longitude pair

The query searches for all documents that intersect (fit in) a georectangle made by the two coordinates `top_left` and `bottom_right`, as in figure 12.4. The user can provide these two coordinates so that we can construct a rectangular shape. Restaurants inside this rectangle are returned as search results, and the rest are dropped.

> **NOTE** We can also represent the vertices of the bounded rectangle as `top_right` and `bottom_left` (as opposed to `top_left` and `bottom_right`). Or we can break it down even further by using coordinates named `top`, `left`, `bottom`, and `right`.

### Working with geoshape data

We executed the `geo_bounding_box` query in listing 12.6 on the schema with fields declared as `geo_point` data types. But can we use the same query if our document consists of a field defined as `geo_shape`, instead? We learned in section 12.3.3 that geospatial data can also be represented using a `geo_shape` data type, remember?

We can use the query from listing 12.6 for geoshape data, except we must swap the URL pointing to the appropriate index. For example, with the `cafes` index and geoshape data at hand, all we need to do is construct the same `geo_bounding_box` query but change the URL to reflect the geoshapes index (`cafes`):

```
GET cafes/_search
{
  "query": {
    "geo_bounding_box": {
      "address": {
        "top_left": {
          "lat": 52,
          "lon": 0.04
        },
        "bottom_right": {
```

The cafes index with geo_shape data type fields

The address field is defined as a geo_shape data type.